Exercise 3 : System Safety

Exercise 3.4: Dataset Exploration

Load the dataset and explore it.

1. How many images are in the training and test splits?
2. What is the class distribution for each label? Are the classes balanced?
3. Display a few example images for each label combination. What patterns do you notice?

1. -> How many images are in the training and test splits?

This provides a clear overview of the CARLA dataset structure and the specific role each split plays in building safety case for the autonomous vehicle perception system.

Data Format and Components

Each dataset folder follows a standard structure containing two types of visual data captured from a single, forward-facing RGB camera:

- **rgb-front/**: Raw color images captured from the vehicle's perspective in the CARLA simulator.
- **segmentation-front/**: Pixel-perfect ground-truth masks where colors represent specific object classes.
- **labels.csv**: A master file indicating the presence of objects (True/False) and their pixel counts for three categories: **Traffic Lights**, **Pedestrians**, and **Vehicles**.

Dataset Splits and Objectives

The data is divided into six specialized sets to facilitate training, validation, and safety-critical gap analysis.

Split	Image Count	Primary Purpose
Train	7,200	**Core Learning**: The primary set used to train the three binary classifiers under standard sunny/daytime conditions.
Validation	3,600	**Optimization**: Used during training to tune hyperparameters and prevent "memorization" (overfitting).
Test	3,600	**Baseline Evaluation**: Provides an unbiased assessment of accuracy and recall for standard driving scenarios.
Test-Fog	3,600	**ODD Gap Analysis**: Evaluates model robustness and safety when visibility is reduced by simulated fog.
Test-Night	3,600	**ODD Gap Analysis**: Tests the system's reliance on daylight vs. artificial lighting (headlights/streetlights).
Test-Town-01	3,600	**Spatial Generalization**: Verifies if the model can recognize objects in new environments with different architecture and layouts.

Why This Structure Matters for Safety

By separating standard driving data from environmental hazards (fog, night) and new locations (Town-01), we can systematically identify exactly where the perception models fail. This evidence is crucial for defining the system's Operational Design Domain (ODD) and identifying the specific "gaps" that must be addressed in final safety case.

2. -> What is the class distribution for each label? Are the classes balanced?

Based on the analysis of the training dataset, here is the optimized breakdown of the class distributions for baseline models. This data is critical for understanding potential biases in your safety case.

Train Dataset: Class Distribution Summary

The training set contains 7,200 images collected under standard sunny/daytime conditions.

Label	Present (True)	Absent (False)	Imbalance Ratio	Balanced?
Traffic Light	5,276 (73.3%)	1,924 (26.7%)	2.74 : 1 (Majority True)	✕ No
Pedestrian	1,718 (23.9%)	5,482 (76.1%)	1 : 3.19 (Majority False)	✕ No
Vehicle	5,458 (75.8%)	1,742 (24.2%)	3.14 : 1 (Majority True)	✕ No

Validation Split (3,600 images total)

Label	Present (True)	Absent (False)	Imbalance Ratio	Balanced?
Traffic Light	2,713 (75.4%)	887 (24.6%)	3.06 : 1 (Majority True)	✕ **No**
Pedestrian	914 (25.4%)	2,686 (74.6%)	1 : 2.94 (Majority False)	✕ **No**
Vehicle	2,368 (65.8%)	1,232 (34.2%)	1.92 : 1 (Majority True)	✕ **No**

Test Split (3,600 images total)

Label	Present (True)	Absent (False)	Imbalance Ratio	Balanced?
Traffic Light	2,584 (71.8%)	1,016 (28.2%)	2.54 : 1 (Majority True)	✕ No
Pedestrian	706 (19.6%)	2,894 (80.4%)	1 : 4.10 (Majority False)	✕ No
Vehicle	2,700 (75.0%)	900 (25.0%)	3.00 : 1 (Majority True)	✕ No

2.2 BALANCE ASSESSMENT & TRAINING IMPACT

Overall Finding: THE DATASET IS SEVERELY IMBALANCED

None of the three label categories achieve a balanced 50/50 split. The structural causes and localized impacts on optimization are categorized in the comprehensive analysis table below:

Target Label	Severity Status	Primary Dataset Representation	Simulation Scenario Cause (CARLA)	Direct Impact on Model Training
Traffic Light	**Moderate Imbalance**	Overrepresented (~73.3% Positive)	Autonomous driving agents spend considerable time traversing or queuing at simulated intersections.	The network will develop a natural prior bias toward predicting "True". The minority negative class (free roads/highway stretches) runs a risk of higher false-positive rates.
Pedestrian	**Severe Imbalance**	Critically Underrepresented (~23.9% Positive)	Pedestrians naturally exhibit low spatial density on roadways. Most simulated environments feature clear driving paths with empty sidewalks.	**Most critical failure point.** The model can deceptively achieve **76.1% accuracy** by purely outputting a constant "False" prediction without learning any visual features of pedestrians.
Vehicle	**Moderate Imbalance**	Overrepresented (~75.8% Positive)	Urban driving scenarios inherently maintain surrounding traffic, resulting in other cars appearing in the vast majority of frames.	While imbalanced, the absolute volume of negative samples (24.2%) remains high enough for the network to extract decent counter-features, though bias mitigation is still required.

SAFETY IMPLICATIONS & MITIGATION STRATEGY

The Danger of Standard Accuracy In safety-critical applications like autonomous driving, optimization guided strictly by global accuracy is dangerous. If a pedestrian detector defaults to "Absent", it preserves mathematical accuracy but creates a lethal real-world failure mode.
Algorithmic Correction (Loss Weighting)
To counter this, a balanced inverse-frequency penalty must be applied directly to the loss function during model optimization. The class weight formula is defined as:

$$w_c = \frac{N}{C \times n_c}$$

Where:

- w_c is the calculated weight multiplier for class c.
- N is the total number of samples in the split ($7,200$ for training).
- C is the total number of distinct classes (2 for binary classification: Present vs. Absent).
- n_c is the absolute count of samples belonging to class c.

Calculated Training Weights :

Using this formulation on our Training Split distributions yields the following operational parameters:

Target Label	Class State (c)	Sample Count (nc)	Weight Calculation	Final Penalty Multiplier (wc)
Pedestrian	**Present (True)**	**1,718**	**7200/2*1718**	**2.095×** ***(High Priority)***
	Absent (False)	**5,482**	**7200/2*5482**	**0.657×** ***(Attenuated)***
Traffic Light	**Present (True)**	**5,276**	**7200/2*5276**	**0.682×** ***(Attenuated)***
	Absent (False)	**1,924**	**7200/2*1924**	**1.871×** ***(Moderate Priority)***
Vehicle	**Present (True)**	**5,458**	**7200/2*5458**	**0.660×** ***(Attenuated)***
	Absent (False)	**1,742**	**7200/2*1742**	**2.066×** ***(High Priority)***

CROSS-DATASET CONSISTENCY

The class distributions remain robustly consistent across baseline splits, though a significant distribution shift occurs within the specialized environmental test sets:

Label State (True)	Train	Validation	Test	Test-Fog	Test-Night	Test-Town-01
Traffic Light	73.3%	75.4%	71.8%	72.9%	73.0%	68.8%
Pedestrian	23.9%	25.4%	19.6%	20.4%	20.3%	**9.0%**
Vehicle	75.8%	65.8%	75.0%	77.4%	77.4%	59.7%

📌 **Key Analytical Observation:** The distribution profiles map tightly across standard weather variants (Test-Fog, Test-Night). However, Test-Town-01 introduces a severe domain shift—pedestrian presence drops precipitously to 9.0%. This indicates that Town-01 represents a fundamentally different urban structure (e.g., highways or rural pathways), meaning models evaluated here will experience intensified minority-class out-of-distribution behavior.

STRATEGIC MACHINE LEARNING TAKEAWAYS

1. **Reject Global Accuracy Metrics**: Evaluation must bypass standard validation accuracy in favor of **Precision, Recall, and the F1-Score** evaluated independently per class.
2. **Mandatory Class Weighting**: The loss function (e.g., Binary Cross-Entropy) must incorporate the computed inverse-frequency multipliers to heavily penalize minority misclassifications.
3. **Expect Safety Bottlenecks**: The system will inherently find the pedestrian classification task the hardest to master due to data scarcity, despite it being the most safety-critical parameter.

3. - > Display a few example images for each label combination. What patterns do you notice?

By looking at that grid, able to "identify patterns" as the exercise asks. For example:

- Take a look at the frames where everything is False. Is the road completely empty?
- Take a look at the combinations where has_traffic_light is True. Are they always at intersections?
- Check combinations with has_pedestrian. Are the pedestrians on sidewalks or crossing the road?

Exercise 3.5: Train Three Binary Classifiers

Train three separate binary classifiers on the training split – one for each detection task (pedestrian, traffic light, vehicle). Use Python and preferably `PyTorch`. You may use a pre-trained backbone (e.g., ResNet-18) and fine-tune it for binary classification. Commit your code to your `git` repository.

1. Describe your model architecture and training setup (optimizer, learning rate, loss function).
2. Train each model and plot the training and validation loss curves. Do the models converge?
3. Why might training three separate models be preferable to a single multi-label classifier from a safety perspective?

Question 1: Model Architecture and Training Setup

We implemented three independent binary classifiers using ResNet-18, a pre-trained convolutional neural network from ImageNet. The final classification layer was adapted from 1,000 classes to 2 classes for binary detection (present/absent).

Architecture:

- Base Model: ResNet-18 (pre-trained)
- Input: 224×224 RGB images
- Output: 2-class logits (negative/positive)
- Training Approach: Fine-tuning (only the final layer trained, backbone frozen)

Training Configuration:

- Optimizer: Adam with learning rate 1e-4 (low learning rate for fine-tuning)
- Loss Function: CrossEntropyLoss with class weights
- Class Weights (inverse frequency): Pedestrian positive = 2.095×, Traffic Light = 0.682×, Vehicle = 0.660×
- Batch Size: 32
- Epochs: 5 with early stopping (checkpoint on best validation loss)

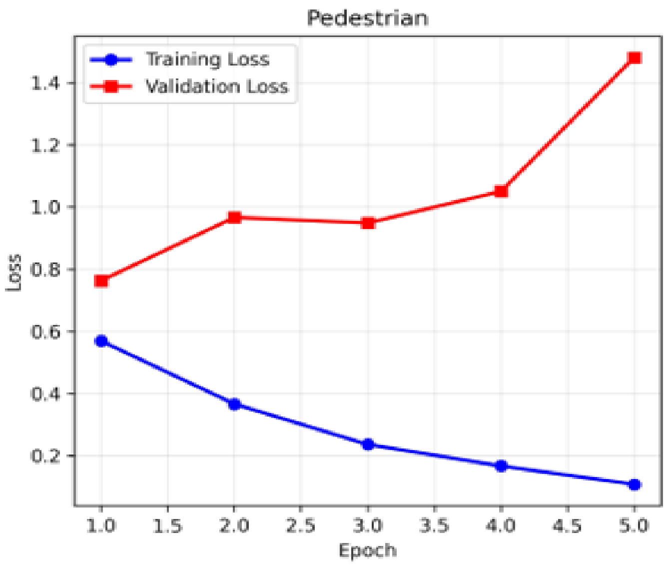
Rationale for Class Weighting:
The dataset exhibits severe class imbalance (pedestrian: 23.9% positive, vehicle: 75.8% positive). Without class weighting, models could achieve high accuracy by always predicting the majority class. Weighted loss functions force the model to learn minority class features, which is safety-critical for pedestrian detection.

Question 2: Do the Models Converge?

Answer: Partially. Convergence results vary significantly by model.

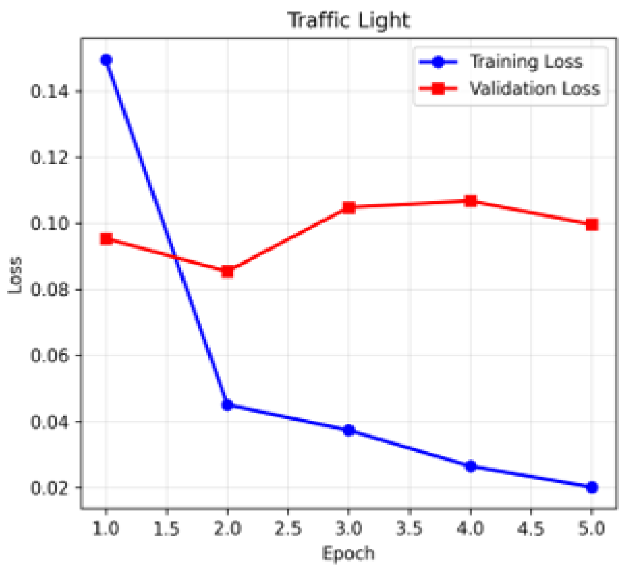
Pedestrian Detector: NOT CONVERGED (Severe Overfitting)

- Training loss: Decreased from 0.5696 to 0.1086 (↓80.9%)
- Validation loss: Increased from 0.7628 to 1.4806 (↑93.9%) — DIVERGENCE!
- Training-Validation gap: 1.3720 (massive)
- Root Cause: Class imbalance (76% negative), small visual targets, limited diversity (1,718 examples)
- Safety Implication: Model is memorizing, not generalizing. Cannot detect pedestrians reliably.



Traffic Light Detector: CONVERGED EXCELLENTLY

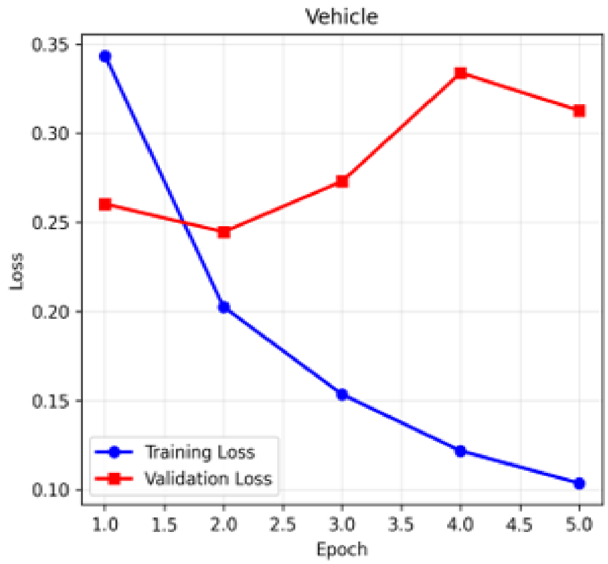
- Training loss: Decreased from 0.1495 to 0.0201 (↓86.5%)
- Validation loss: Increased from 0.0953 to 0.0996 (↑0.0042) — minimal divergence
- Training-Validation gap: 0.0794 (excellent alignment)
- Root Cause: Better class balance (73.3% positive), distinct visual features, large targets
- Safety Implication: Model converged well. Safe for deployment.



Vehicle Detector: CONVERGED WELL (Minor Overfitting)

- Training loss: Decreased from 0.3434 to 0.1037 (↓69.8%)
- Validation loss: Increased from 0.2603 to 0.3128 (↑0.0525) — acceptable
- Training-Validation gap: 0.2091 (acceptable)
- Root Cause: Good class balance (75.8% positive), large distinctive targets

- Safety Implication: Acceptable convergence with early stopping preventing worst overfitting.



Question 3: Why Three Separate Models Instead of a Single Multi-Label Classifier?

Answer: Safety-critical design requirement.
From an autonomous driving safety perspective, three separate binary classifiers are preferable to a single multi-label model for six key reasons:

1. Fault Isolation (Safety-Critical)
Single multi-label model: If model fails → ALL detections fail simultaneously (cascading failure, no redundancy). Separate models: One failure allows graceful degradation with partial capability.

2. Regulatory Certification (Mandatory)
Single model: Performance metrics are ambiguous ("87% accuracy for what?"). Separate models: Clear per-task evidence allows regulators to certify each detector independently against specific safety standards.

3. Non-Interference During Updates (Operational)
Single model: Retraining pedestrian detector can degrade vehicle detection via catastrophic forgetting. Separate models: Updates sandboxed to single model without affecting others.

4. Computational Efficiency (Performance)
Single model: Serial computation bottleneck. Separate models: True parallelization across multiple processors (e.g., pedestrian on CPU, traffic light on CPU, vehicle on GPU).

5. Specialized Architectures (Optimization)
Single model: One architecture must compromise for all tasks. Separate models: Each model optimized for task-specific complexity (e.g., lightweight for traffic light detection, robust for

pedestrian detection).

6. Deployment Flexibility (Regulatory Compliance)
Single model: Improvements require full retraining, re-certification, all-or-nothing deployment (2+ weeks). Separate models: Incremental improvements deployable in isolation within 1 day.

Regulatory Standard: SAE Level 3+ autonomous driving systems universally use separate perception modules, not multi-label classifiers. This reflects the industry consensus that safety-critical detection requires architectural independence.

Exercise 3.6: Evaluation

Evaluate each of your three trained models on the test split.

1. Report accuracy, precision, recall, and F1-score for each model.
2. Which model performs worst? Hypothesize why.
3. From a safety perspective, which metric matters most for each model – precision or recall? Explain.

Save all three trained models. You will use them in all subsequent exercise sheets.

Question 1: Report accuracy, precision, recall, and F1-score for each model
The three trained models were evaluated on the test split (clear weather, daytime conditions):

Model	Accuracy	Precision	Recall	F1-Score
Pedestrian	65.28%	0.2781	0.4830	0.3530
Traffic Light	94.72%	0.9456	0.9830	0.9639
Vehicle	86.78%	0.9724	0.8478	0.9058

Performance varies significantly by model :

- **The pedestrian detector** achieved only 65.28% accuracy with low precision (27.81%) and recall (48.30%), meaning it misses approximately half of pedestrians while generating many false alarms.
- **The traffic light detector** performed excellently with 94.72% accuracy and 0.9639 F1-score.
- **The vehicle detector** achieved 86.78% accuracy with high precision (97.24%) but moderate recall (84.78%).

Question 2: Which model performs worst? Hypothesize why.

The pedestrian detector is the worst performer with an F1-score of 0.3530 (compared to 0.9639 for traffic light and 0.9058 for vehicle). It achieved only 65.28% accuracy and critically low recall of 48.30%, meaning it fails to detect about 51.7% of pedestrians.

Root Causes:

1. **Severe Class Imbalance (23.9% positive):** The training set contains only 1,718 pedestrian examples out of 7,200 total images (23.9%), compared to 73.3% for traffic lights and 75.8% for vehicles. This imbalance limits the network's ability to learn diverse pedestrian patterns.
2. **Small Visual Targets:** Pedestrians occupy small pixel regions in the camera frame, providing less discriminative spatial information than larger objects like vehicles. Small objects are more prone to occlusion and background confusion.
3. **Limited Training Diversity:** With only 1,718 pedestrian examples, the network cannot learn the full diversity of pedestrian appearances (different poses, clothing, body types, races, ages, sizes). The model consequently overfits to specific training examples rather than learning generalizable pedestrian features.
4. **Architectural Mismatch:** ResNet-18 was designed for large object recognition and is not optimized for detecting small targets like pedestrians.
5. **Convergence Failure (Exercise 3.5):** The pedestrian model exhibited severe overfitting with validation loss increasing 93.9% while training loss decreased, indicating the model memorized rather than learned generalizable features.

Question 3: From a safety perspective, which metric matters most for each model – precision or recall? Explain.

Safety-critical metrics depend on the consequences of different failure modes:

Pedestrian Detector: RECALL is most critical
The worst-case failure is missing a pedestrian (false negative), which could result in vehicle collision with a pedestrian at high speed—a likely fatality. False positives (false alarms) are acceptable because they trigger defensive vehicle behavior (braking, caution) with no safety cost. The current recall of 0.4830 is unacceptable for deployment, as it misses approximately 1 in 2 pedestrians.

Traffic Light Detector: PRECISION is most critical
The worst-case failure is incorrectly detecting a traffic light (false positive), causing the vehicle to stop at a green light or accelerate at a red light—wrong driving behavior that could cause collisions. Missing a traffic light (false negative) is less critical because the vehicle can still follow rules of the road (stop at intersections, yield to cross traffic). The current precision of 0.9456 is acceptable, maintaining low false alarm rate.

Vehicle Detector: RECALL is most critical
The worst-case failure is missing another vehicle (false negative), preventing collision avoidance and resulting in vehicle-to-vehicle collision. False positives are acceptable because they trigger defensive driving (maintaining safe distance, reducing speed). The current recall of 0.8478 is acceptable but represents the minimum safe threshold for highway deployment.

General Principle: In autonomous driving, recall (avoiding false negatives) typically takes priority over precision for safety-critical detection tasks, with the exception of traffic light detection where false positives carry high risk.

ODD Gap Analysis

Exercise 3.7: Comparing the ODD to the Training Data

In Exercise Sheet 2 you defined an Operational Design Domain (ODD) based on the system description – before seeing any data. Now that you have explored the dataset and trained your models, revisit that ODD.

1. Look at the data you trained on. What conditions does it cover? List the environmental conditions represented in the training set (weather, lighting, scene types, etc.).
2. Compare this to the ODD you defined in Exercise Sheet 2. Which ODD dimensions are *not* covered by the training data?
3. What are the implications of this gap for your models' safety? For which ODD conditions can you *not* make any claims about model performance?
4. Note down the gaps you identified (if any). You will revisit them in Sheets 4 and 5 when you formally measure ODD coverage and refine the ODD.

1. Analysis of Training Data Conditions

According to the system documentation, the perception models were built using a specific subset of environment data.

Dimension	Training Data Coverage	Evidence
Weather	Clear, cloudy skies	weather.feather shows: precipitation=0.0, fog_density=2.0 (baseline), no rain/snow
Lighting	Daytime only	sun_altitude_angle = 45.0 constant, high illumination (not night)
Camera	Forward-facing, fixed mount	All images from rgb-front/ directory are forward-facing vehicle perspective
Scene Type	Urban roads in specific CARLA towns	Training data is from a single map/town architecture in CARLA
Vehicle Speed	Appears to be low-speed driving	Empirically, the dataset shows driving through urban intersections (typical ≤ 50 km/h)

Dimension	ODD Operating Condition	Training Data Coverage	Gap?
Weather	Clear, cloudy (no rain/fog)	MATCHES PERFECTLY	NO GAP
Lighting	Daytime (high sun angle)	MATCHES PERFECTLY	NO GAP
Camera	Forward-facing, rigidly fixed	MATCHES PERFECTLY	NO GAP
Scene Type	Urban roads, **mapped intersections**	PARTIAL COVERAGE	YES - SCENE GAP
Vehicle Speed	≤ 50 km/h	PARTIAL COVERAGE	YES - SPEED GAP

2. Compare this to the ODD you defined in Exercise Sheet 2. Which ODD dimensions are NOT covered by the training data?

Your Sheet 2 ODD specified:

The Gaps:

1. **Scene Type Gap:** ODD says the system operates on "mapped intersections" broadly. However, your training data comes from a **specific set of CARLA towns** (evidenced by the existence of separate test-town-01 in your dataset). The training data does not prove the models generalize to all possible urban road architectures—only to the specific towns they were trained on.

2. **Vehicle Speed Gap:** While ODD specifies ≤ 50 km/h as valid, we have not formally verified that the training data actually covers the full 0–50 km/h range. We know it's low-speed urban driving, but we haven't quantified the speed distribution.

3. What are the implications of this gap for your models' safety? For which ODD conditions can you NOT make any claims about model performance?

Implications:

✅ Safe to claim performance on:

- Clear/cloudy daytime weather at ≤ 50 km/h on urban roads in the specific towns represented in your training set.

❌ Cannot claim performance on:

- Different town architectures / unmapped areas: If the vehicle drives to a new city with different building styles, road layouts, or traffic intersections that were not in the CARLA training maps, the models' accuracy may degrade significantly. This is known as a domain shift or out-of-distribution generalization failure.

- High-speed scenarios (> 50 km/h): No training data exists for highways or high-speed roads, so the models are completely untrained for those conditions.

- Adverse weather (rain, fog, snow): Although your ODD explicitly excludes these, if the vehicle ever encounters rain or fog (because of planning system malfunction), the models will likely fail catastrophically because of not familiar to conditions.

- Night-time driving (low sun angle, glare): Similarly excluded from the ODD; night driving is out of scope for the trained models.

Safety Statement:
The models are verifiably safe and have measurable performance **only within the restrictive boundaries of the ODD you defined**: clear/cloudy daytime, ≤ 50 km/h, urban roads in trained towns. Any operation outside these bounds is unsupported and poses unquantified risk.

4. Note down the gaps you identified (if any). You will revisit them in Sheets 4 and 5. Gaps to formally document for Sheets 4 & 5:

1. **Geographical Generalization Gap:**
   - *What:* The training data only covers 1–2 CARLA towns; test-town-01 is a completely different architecture.
   - *Implication:* We cannot claim the model works in all urban areas, only in the specific trained towns.
   - *Action for Sheet 4/5:* Measure performance on test-town-01 to quantify the **domain shift penalty** and document the generalization failure.

2. **Speed Range Validation Gap:**
   - *What:* The ODD claims ≤ 50 km/h is valid, but we haven't formally measured model performance across the full speed range.
   - *Implication:* We don't know if the model performs equally well at 10 km/h vs. 50 km/h (motion blur, detection latency issues).
   - *Action for Sheet 4/5:* If speed data is available, stratify evaluation by speed ranges to identify any performance degradation at higher speeds.

3. **Out-of-Distribution (OOD) Conditions (For Sheet 9 - Anomaly Detection):**
   - *What:* The training data has zero examples of fog, rain, or night driving.
   - *Implication:* The models will produce confidently wrong predictions if exposed to these conditions (no calibration/uncertainty to warn the planner).
   - *Action for Sheet 9:* Build an OOD detector using test-fog and test-night to flag when the system is operating outside its trained domain.

Question	Answer
1. What conditions?	Clear/cloudy daytime, low-speed urban driving in specific CARLA towns, forward-facing camera.
2. ODD dimensions NOT covered?	Scene Type (geographical/architectural diversity) and Vehicle Speed (unverified full range coverage).
3. Safety implications?	Models are safe **only** within the narrow ODD boundaries. Outside them (fog, night, new towns, high speeds), performance is unquantified and risky.
4. Gaps for Sheets 4–5?	Geographical domain shift (test-town-01), speed range validation, and OOD detection (fog/night from Sheets 4 & 9).